

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Scenario Based Questions

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Scenario Based Questions	Assessment:	Assessment Tool 1– Scenario Based Questions
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	A. Reddy Prasad	Reg. No.:	192472045
Section / Batch:	2024	Date:	20-08-2026

Code of Conduct

I A. Reddy Prasad (192472045) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A. Reddy Prasad

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	

		analyze the case deeply				
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Question 6 – Confidence Interval

Problem:

An online learning platform measures the completion time of a course for 100 students. The mean completion time is 14 days and the standard deviation is 3 days. Construct a 95% confidence interval for the true mean completion time.

Given:

$$n = 100$$

$$\bar{x} = 14 \text{ days}$$

$$s = 3 \text{ days}$$

$$\text{Confidence level} = 95\%$$

$$Z = 1.96$$

Formula:

$$\text{Confidence Interval} = \bar{x} \pm Z \times (s / \sqrt{n})$$

Calculation:

$$\text{Standard Error (SE)} = s / \sqrt{n}$$

$$SE = 3 / \sqrt{100}$$

$$SE = 3 / 10$$

$$SE = 0.3$$

$$\text{Margin of Error} = 1.96 \times 0.3$$

$$\text{Margin of Error} = 0.588$$

$$\text{Confidence Interval} = 14 \pm 0.588$$

$$\text{Lower Limit} = 14 - 0.588 = 13.412$$

$$\text{Upper Limit} = 14 + 0.588 = 14.588$$

Answer:

95% Confidence Interval = (13.412, 14.588) days

Interpretation:

We are 95% confident that the true mean completion time of the course is between **13.412 days and 14.588 days**.

Question 7 – Confidence Interval for Proportion

Problem:

A hospital surveys 400 randomly selected patients, and 340 report that they are satisfied with the service. Construct a 95% confidence interval for the proportion of satisfied patients.

Given:

$$n = 400$$

$$\text{Number of satisfied patients} = 340$$

$$\text{Confidence level} = 95\%$$

$$Z = 1.96$$

Sample Proportion:

$$\hat{p} = 340 / 400$$

$$\hat{p} = 0.85$$

$$\hat{q} = 1 - \hat{p}$$

$$\hat{q} = 1 - 0.85$$

$$\hat{q} = 0.15$$

Formula:

$$\text{Confidence Interval} = \hat{p} \pm Z \times \sqrt{[\hat{p}(1 - \hat{p}) / n]}$$

Calculation:

$$SE = \sqrt{[(0.85 \times 0.15) / 400]}$$

$$SE = \sqrt{(0.1275 / 400)}$$

$$SE = \sqrt{0.00031875}$$

$$SE \approx 0.01785$$

$$\text{Margin of Error} = 1.96 \times 0.01785$$

$$\text{Margin of Error} \approx 0.03499$$

$$\text{Confidence Interval} = 0.85 \pm 0.03499$$

$$\text{Lower Limit} = 0.85 - 0.03499$$

$$\text{Lower Limit} = 0.81501$$

$$\text{Upper Limit} = 0.85 + 0.03499$$

$$\text{Upper Limit} = 0.88499$$

Answer:

$$\text{95\% Confidence Interval} = (0.815, 0.885)$$

In percentage:

$$\text{95\% Confidence Interval} = (81.5\%, 88.5\%)$$

Interpretation:

We are 95% confident that the true proportion of satisfied patients is between **81.5% and 88.5%**.

Question 8 – 99% Confidence Interval

Problem:

A telecom company wants to estimate the average monthly data usage of its customers. A sample of 225 customers has a mean usage of 16 GB and a standard deviation of 5 GB. Construct a 99% confidence interval and interpret the result.

Given:

$$n = 225$$

$$\bar{x} = 16 \text{ GB}$$

$$s = 5 \text{ GB}$$

$$\text{Confidence level} = 99\%$$

$$Z = 2.576$$

Formula:

$$\text{Confidence Interval} = \bar{x} \pm Z \times (s / \sqrt{n})$$

Calculation:

$$\text{Standard Error (SE)} = 5 / \sqrt{225}$$

$$SE = 5 / 15$$

$$SE = 0.3333$$

$$\text{Margin of Error} = 2.576 \times 0.3333$$

$$\text{Margin of Error} \approx 0.859$$

$$\text{Confidence Interval} = 16 \pm 0.859$$

$$\text{Lower Limit} = 16 - 0.859$$

$$\text{Lower Limit} = 15.141$$

$$\text{Upper Limit} = 16 + 0.859$$

$$\text{Upper Limit} = 16.859$$

Answer:

99% Confidence Interval = (15.141, 16.859) GB

Approximately:

99% Confidence Interval = (15.14, 16.86) GB

Interpretation:

We are 99% confident that the true average monthly data usage of the customers is between **15.14 GB and 16.86 GB**.

Question 9 – Hypothesis Testing Using Confidence Interval

Problem:

A company claims that the average time taken by its customer-support team to resolve a complaint is 24 hours. A sample produces a 95% confidence interval of (25.2, 27.8) hours. Based only on this confidence interval, would you reject the company's claim? Explain.

Given:

Claimed mean = 24 hours

95% Confidence Interval = (25.2, 27.8) hours

Step 1: Check whether the claimed value is inside the confidence interval.

Lower Limit = 25.2 hours

Upper Limit = 27.8 hours

Claimed value = 24 hours

Since:

$$24 < 25.2$$

The value 24 hours is **outside** the confidence interval.

Decision:

Reject the company's claim.

Explanation:

The claimed mean of 24 hours does not lie within the 95% confidence interval of (25.2, 27.8) hours. Therefore, based on the confidence interval, there is sufficient evidence at the 5% significance level to reject the company's claim.

Answer:

The company's claim is rejected because 24 hours is outside the 95% confidence interval.

Question 10 – Hypothesis Testing Using Confidence Interval

Problem:

A manufacturer claims that the average weight of a packaged product is 500 grams. A 95% confidence interval based on a sample is (496, 502) grams. Is the claimed value of 500 grams consistent with the sample evidence? Explain your answer.

Given:

Claimed mean = 500 grams

95% Confidence Interval = (496, 502) grams

Step 1: Check whether the claimed value is inside the confidence interval.

Lower Limit = 496 grams

Upper Limit = 502 grams

Claimed value = 500 grams

Since:

$$496 < 500 < 502$$

The value 500 grams is **inside** the confidence interval.

Decision:

Do not reject the manufacturer's claim.

Explanation:

The claimed mean of 500 grams lies within the 95% confidence interval of (496, 502) grams. Therefore, the sample evidence is consistent with the manufacturer's claim.

Answer:

The manufacturer's claim is consistent with the sample evidence because 500 grams lies within the 95% confidence interval.